

MATLAB APP — ANNOTATED GUIDE

Planar Phased Array Designer

Every control, pointed at and explained on the screen it lives on.

The window — what each area is

Task tabs

Array / Element / Beam / View. Both the ribbon and the left panel change with the tab.

Ribbon

Buttons for the active tab — here, the six aperture outlines.

Control panel

The numeric settings for the active tab. It scrolls.

Compute + Auto

Leave Auto ticked and every change recomputes on its own.

Metrics

Every computed figure for the current design.



Configuration

Design and operating frequency, and the Beam squint switch. Visible on every tab.

File / Export

Save, Load, CSV, TSV. Also on every tab.

Element layout

The aperture to scale. Click to add, select or remove elements.

Per-element table

Amplitude, phase and rotation, editable cell by cell.

3D pattern

Captioned with the steer angle and the gain figures.

Principal cut

One slice, with HPBW / FNBW / sidelobe readouts.

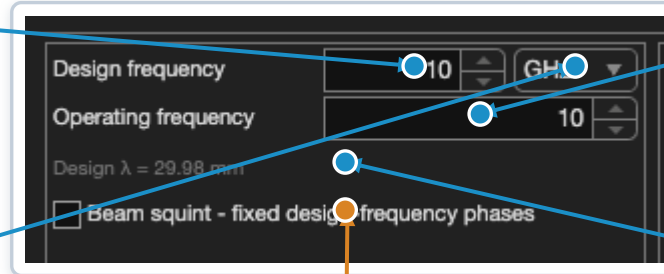
Configuration — the two frequencies and the squint switch

Design frequency

The RULER. Every dx / dy you type is in wavelengths at this frequency. Change it and the physical array rescales — the beam does not move.

GHz / MHz

CONVERTS, it does not reinterpret. 10 GHz becomes 10000 MHz — the same design in different units, with the spinner limits moved to match.



Operating frequency

The frequency the array is actually driven at. Propagation is evaluated here, so the aperture is electrically bigger or smaller than at the design frequency.

Design λ readout

$\lambda = 299.792458 / f(\text{GHz})$ mm. This is the number every physical export is built from.

Beam squint — fixed design-frequency phases

OFF: phases are recomputed at the operating frequency, so the beam holds its commanded angle (retuned / true-time-delay array).
ON: phases stay frozen at the design frequency, so changing the operating frequency walks the beam off the commanded angle.

Rule of thumb

Set the DESIGN frequency first and leave it alone — it defines what "0.5 λ " means and what every exported millimetre is worth. Move the OPERATING frequency afterwards to ask "what happens at a different frequency?".

File and Export — which button writes what

Save → array_config.mat

The whole design: geometry, element table, the imported far-field itself, and every view setting.

Load → reads that .mat

Restores all of it. A config saved with an incompatible imported pattern is refused outright, leaving the current design untouched.



CSV → array_elements.csv

The element table as plain CSV: x, y (in wavelengths), amplitude, phase and rotation. For a spreadsheet or your own scripts.

TSV → array_cst.tsv

The element list in CST's tab-separated array-import layout — and the one export in PHYSICAL units. See the next slide.

Watch the frequency before you press TSV

Everywhere else in the app, element positions are in WAVELENGTHS. The TSV is the point where they become millimetres — and the number they are multiplied by comes from the Design frequency alone.

$X(\text{metres}) = x(\text{wavelengths}) \times \lambda_{\text{design}}$, $\lambda_{\text{design}} = 299.792458 / f_{\text{design}}(\text{GHz})$ mm

So a 0.5λ element at a 10 GHz design frequency is written as 0.0149896 m (14.99 mm). Set the design frequency to 20 GHz and the SAME array exports at 7.49 mm — half the size — while nothing on screen appears to have changed. The operating frequency has no effect on the file at all.

Inside the TSV file

Matched to a real CST 2025 export, not a guessed format. The header block is "#" -prefixed and the rows are tab-separated.

```
# Created by MATLAB phasedArrayDesigner.m (not CST Studio Suite)
# On 10-Sep-2026 21:13:11
# unit: meters
# design frequency: 10 GHz
# phase reference frequency: 12 GHz
# operating frequency: 12 GHz
# beam squint: OFF (phases retuned at operating frequency)
# Element      X          Y  Z  Magnitude  Phase  Phi  Theta  Gamma
1,1            0          0  0  1          0    0    0    0
2,1            0.0149896229 0  0  1        252    0    0    0
3,1            0.0299792458 0  0  1        144    0    0    0
```

a real export — 1 x 3 at 0.5 λ, steered 30°. Columns aligned here for reading; the file is tab-separated.

Before exporting, check

- Positions always use the DESIGN frequency. Phases use whichever frequency the header's "phase reference" line names.
- Those two can differ. With Beam squint ON the phases stay at the design frequency while the array runs at another one.
- The unit selector is only a display choice — it does not change the file.

Element	"col,row" — the first number tracks X, the second tracks Y.
X Y Z	Position in METRES. Z is always 0 — the array is coplanar.
Magnitude	Amplitude after the taper. A negative amplitude is written positive with +180° folded into Phase.
Phase	The TOTAL applied feed phase: steering + your manual offset (+ rotation if that box is ticked). NOT the table's editable Phase column, which is the offset alone.
Phi	This app's per-element Rotation.
Theta / Gamma	Always 0. A coplanar array has no out-of-plane tilt.

Array tab — geometry

Placement mode

Uniform grid fills every cell. Sparse lets you click cells. Sub-position offers 9 offsets inside a cell.

Rows M / Cols N

Element count of the lattice, 1–64 each.

dx / dy (λ)

Spacing in DESIGN wavelengths, 0.05–5. Half a wavelength is the usual choice.

Array shape

Masks the grid to an outline. Same setting as the six ribbon buttons.

The screenshot shows the 'Array tab — geometry' control panel. It features several settings with callout lines pointing to them from external text boxes:

- Placement mode:** A dropdown menu set to 'Uniform grid'.
- Rows M / Cols N:** Two input fields, both set to '8'.
- dx / dy (λ):** Two input fields, set to '0.5' and '0.5' respectively.
- Array shape:** A dropdown menu set to 'Custom'.
- Grid angle (deg):** An input field set to '90'.
- Row stagger (λ):** An input field set to '0'.
- Stagger angle (deg):** An input field set to '0'.
- Rotation (0=+y, +=CW):** An input field set to '45'.
- Apply to selected:** A set of buttons including a radio button, a minus sign, a plus sign, and a zero.

Grid angle (deg)

Skews the lattice. 90° rectangular, 60° triangular.

Row stagger (λ)

Offsets alternate rows sideways.

Stagger angle (deg)

The direction the stagger runs along.

Rotation (0=+y, +=CW)

Physical rotation of an element about its own axis.

Apply to selected

Set / - / + / 0 — apply, step or clear rotation on the selection.

Array tab — rotation, moving, sequential rotation

Seq. rot. step (deg)

The step used by "Apply physical sequential rotation to all". 90° gives the usual 0/90/180/270.

Rot. block (rows × cols)

The repeating block that the rotation sequence cycles over.

The screenshot shows a control panel for an array. It has a dark background with white text and controls. The controls are arranged in a vertical stack. The first section has a label 'Seq. rot. step (deg)' with a value of 90. Below it is a label 'Rot. block (rows x cols)' with a value of 2x2. Then there is a checkbox labeled 'Sequential rot.' with a sub-label 'extra rot angle -> feed ...'. Below this is a button 'Apply physical sequential rotation to all'. The next section has a label 'Move dx / dy (λ)' with a value of 1. Below this are three buttons: 'Move selected by (dx,dy)', 'Reset selected to lattice', and 'Delete selected element(s)'. Then there is a button 'Clear all elements'. The final section has a button 'Compute pattern' and a checkbox 'Auto' which is checked. Blue circles and lines point from the explanatory text boxes to these specific controls.

extra rot angle → feed phase

Also add each element's rotation to its feed phase. This is what turns a mechanically rotated array into a circularly polarised one.

Move dx / dy (λ)

How far "Move selected by (dx,dy)" shifts the selected elements.

Compute pattern

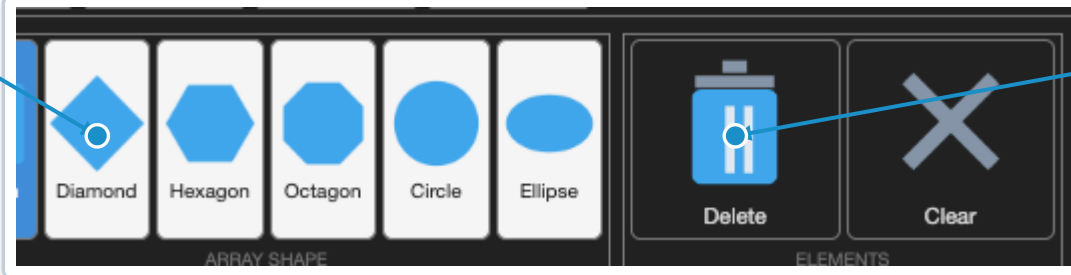
Recomputes everything. With Auto ticked you rarely need it.

Outline shapes and how elements get placed

Six outline presets mask a rectangular grid. The ribbon buttons and the Array shape dropdown are one setting seen twice.

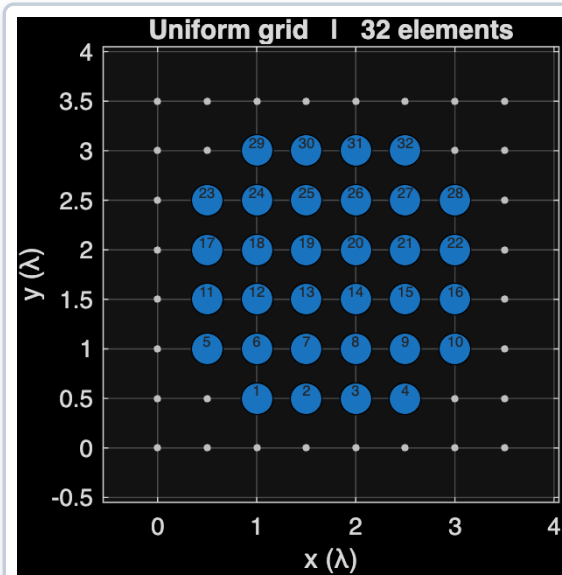
Shape buttons

Custom · Diamond · Hexagon · Octagon · Circle · Ellipse. Clicking one also moves the dropdown.

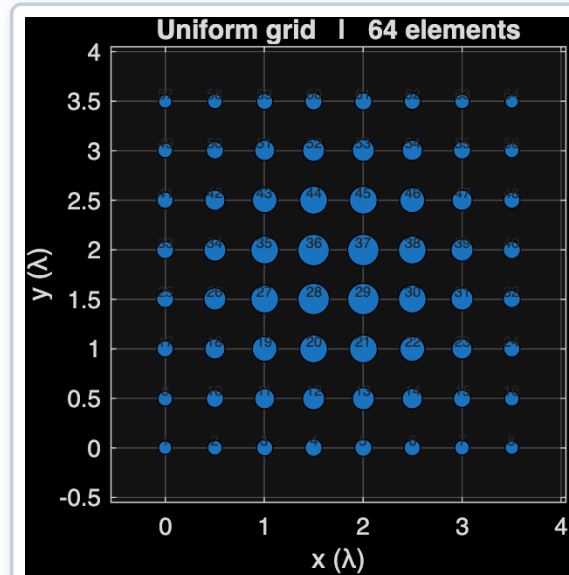


Delete / Clear

Delete removes the selected elements; Clear empties the array. A shape that would leave a row empty is refused instead of applied.



Circle mask on 8×8 — 32 elements



Marker size = element amplitude

THE THREE PLACEMENT MODES

- Uniform grid — every cell of the $M \times N$ lattice is populated.
- Sparse (click cells) — click a cell to add an element, click an element to select it, Delete to remove.
- Sub-position (9-pt) — click a cell then pick one of nine offsets inside it, including the diagonals. A cell can hold several, so repeated clicks densify the array.

The element layout — clicking the aperture

Title line

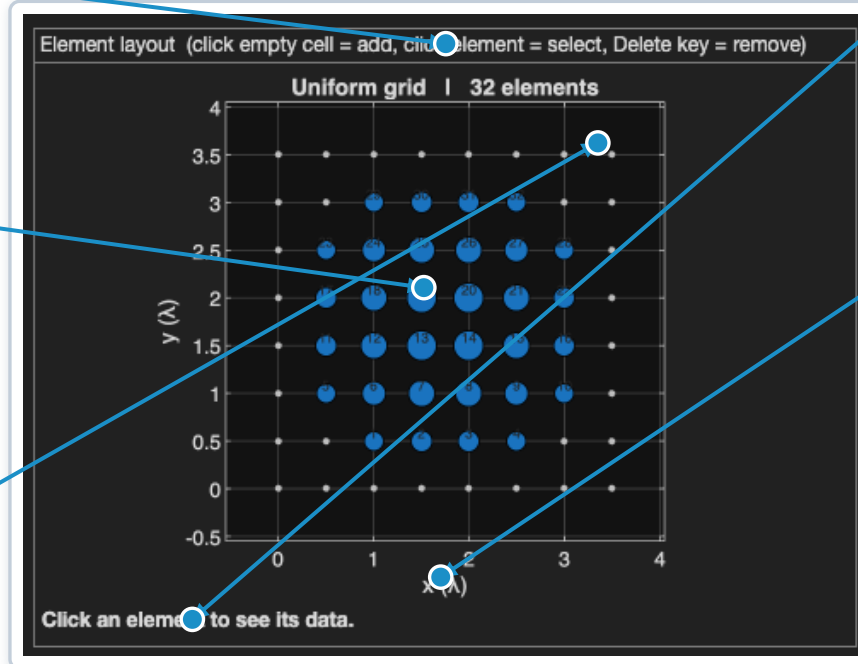
Placement mode and the live element count. It updates as you add or mask elements.

Filled markers = elements

Drawn to scale in wavelengths. Marker SIZE is the element's amplitude — the Chebyshev taper here is visible directly on the aperture.

Faint dots = empty lattice

The full $M \times N$ lattice is always drawn behind, so the cells this Circle mask trimmed away stay visible as empty dots.



Click to edit

In Sparse or Sub-position mode, clicking an empty cell adds an element and clicking a placed one selects it. The Delete key removes the selection.

Axes are wavelengths

x and y in DESIGN wavelengths — the same units as dx / dy , not millimetres.

Selection drives the ribbon

"Apply to selected", "Move selected by (dx, dy) ", "Reset selected to lattice" and Delete all act on whatever is selected here or in the table.

The per-element table — editing one element at a time

x / y (λ)

Position in design wavelengths. Read-only here — move elements with the ribbon.

Amp

Element amplitude. Typing here only survives if Amplitude taper is set to "Manual (table)".

Phase (deg)

Your MANUAL offset only — not the steering phase. Steering is added on top of it, which is why this column does not move when you steer.

Per-element data (edit Amp / Phase / Rot directly)

	x (λ)	y (λ)	Amp	Phase (d...	Rot (deg)	Feed pl
1	0	0	1.0000	0	0	
2	0.5000	0	1.0000	0	0	
3	1.0000	0	1.0000	0	0	
4	1.5000	0	1.0000	0	0	
5	0	0.5000	1.0000	0	0	
6	0.5000	0.5000	1.0000	0	0	
7	1.0000	0.5000	1.0000	0	0	
8	1.5000	0.5000	1.0000	0	0	
9	0	1.0000	1.0000	0	0	

Rot (deg)

Physical rotation of that element. Only reaches the feed phase if the "extra rot angle" box is ticked.

Feed phase

The TOTAL applied phase: steering + manual offset (+ rotation when enabled). This is the number the TSV writes and the phase map shows.

Clicking a row selects that element on the layout too — the two views share one selection.

Element tab — what one element radiates

Element factor

Isotropic · $\cos^q(\theta)$ · Dipole (linear pol) · Patch ($\cos^q \times \text{lin pol}$) · Custom (formula) · Imported (CST far-field). The array factor alone assumes point sources; this is what makes the answer physical.

q exponent

The q in $\cos^q(\theta)$, 0–6. Directivity of a $\cos^q$ element is $D = 2(q+1)$.

Custom E_theta / E_phi

Your own formulas in MATLAB syntax; th , ph and ct are available. Set E_{ϕ} to 0 for a purely theta-polarised element.

The screenshot shows the 'Element tab' control panel with the following fields and callouts:

- Element factor:** A dropdown menu set to 'Isotropic'. A callout points to this dropdown.
- q exponent:** A numeric input field set to '1.5'. A callout points to this field.
- Custom E_theta(th,ph):** A text input field containing 'ct.^2'. A callout points to this field.
- Custom E_phi(th,ph):** A text input field containing '0'. A callout points to this field.
- vars:** A small text block below the custom fields explaining variables: `th, ph, ct, st` (where `th` and `ph` also work). It includes a note: `NOTE ct = max(cosd(th), 0), clamped at 0 -- for back radiation in full-sphere mode use cosd(th) instead. Use dot operators (.* and .^). E_phi=0 gives a linear-pol pattern; set it nonzero for elliptical/CP custom elements. Bad formula -> Isotropic`.
- Load far-field pattern (.txt)...:** A button with a callout pointing to it.
- Imported unit cell (embedded element):** An unchecked checkbox with a callout pointing to it.
- Total efficiency (%):** A numeric input field set to '100'. A callout points to this field.

Total efficiency (%)

Radiation efficiency. Used to turn an imported REALIZED GAIN into a directivity: $D = RG - 10 \cdot \log_{10}(\text{eff})$.

Load far-field pattern (.txt)

Imports a CST Farfield ASCII export. On success the Element factor switches itself to "Imported" and the caption shows the file's realized-gain peak.

Imported unit cell

Ticked = the file is a periodic unit cell, so its peak IS the embedded element's realized gain and the array figures scale from that gain. Unticked = an isolated element, reported as a directivity.

Importing a CST far-field — what is accepted

- 1 In CST, export the far-field as Farfield ASCII with REALIZED GAIN in dBi and Theta/Phi components — the header must carry Abs(Grlz), Abs(Theta), Phase(Theta), Abs(Phi), Phase(Phi).
- 2 Export the full sphere: theta 0...180° and a complete phi circle, regular, with no missing angles.
- 3 Element tab → "Load far-field pattern (.txt)..." and pick the file.
- 4 The Element factor switches itself to "Imported (CST far-field)" and the button caption shows the file name and its realized-gain peak.
- 5 Enter the radiation efficiency in Total efficiency (%) so a directivity can be reported as well as a gain.

✓ Accepted

- Realized gain (Grlz) in dBi
- Complex Theta and Phi components
- Complete, regular theta/phi grid
- Optional axial-ratio column

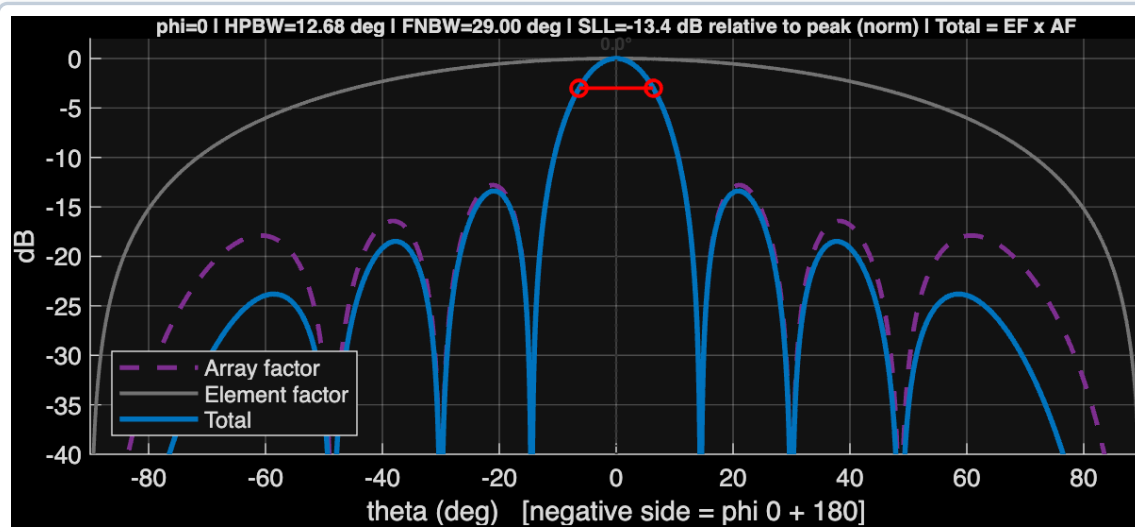
✗ Refused, with a reason

- Directivity or plain gain exports
- Linear E-field (V/m) exports
- Ludwig-3 co/cross-pol
- Folded grids or missing angles

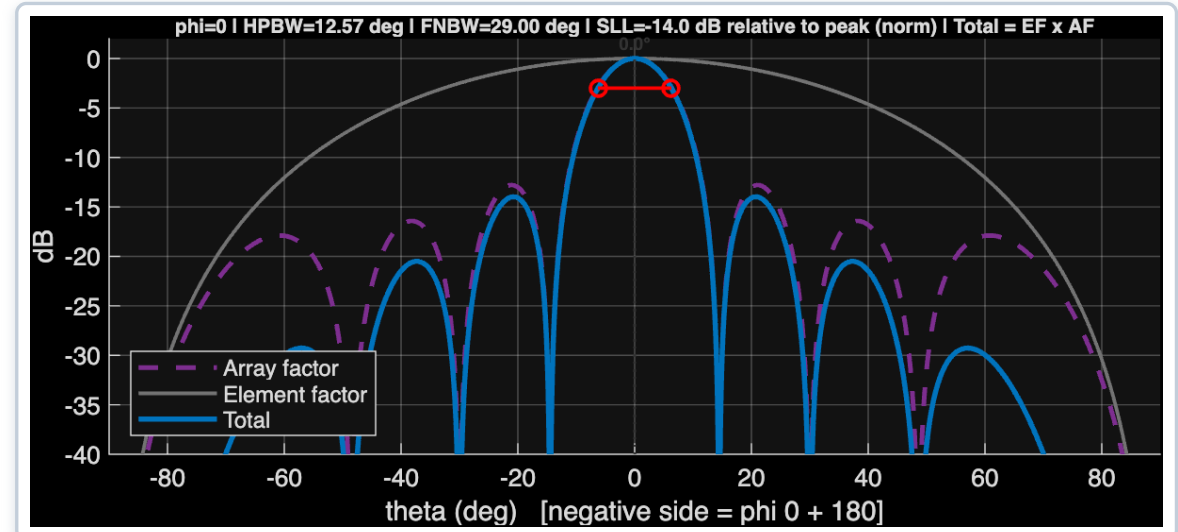
The rule behind all of it: never fill in what was not exported. A pattern with holes would have to be invented, and an invented element is indistinguishable from a measured one once it is in the array.

The element factor changes the whole answer

Same 8×8 aperture, same steering — only the element model changed. Grey is the element factor, blue the total you actually get.



$\cos^2(\theta)$ element — rolls off toward the horizon



Patch element — a different roll-off, and a polarisation

Why it matters for scanning

A directive element falls away as you steer off broadside, so the realised gain drops faster than the array factor alone suggests. That gap is the scan loss the tool reports.

Isolated element or unit cell

With "Imported unit cell" ticked the file's peak IS the embedded element's realized GAIN, coupling and loss included, and array figures scale from that gain rather than from a pattern integral.

Beam tab — pointing, weighting, and the analysis tools

Steer theta / phi

Where you COMMAND the beam. Theta $-90\ldots 90^\circ$, phi $-180\ldots 360^\circ$. Under beam squint this is not where the beam ends up — the cut plot's peak label is.

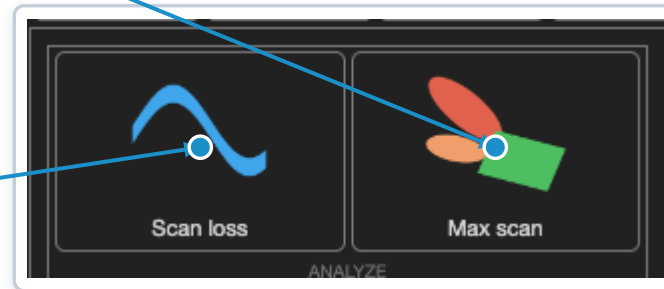


Amplitude taper

Uniform · Hamming · Hanning · Chebyshev · Taylor · Binomial · Manual (table). Chebyshev and Taylor synthesise to a fixed 30 dB design sidelobe level. "Manual" leaves the amplitudes you typed alone.

Max scan

Sweeps the steer angle in 2° steps and stops at the first angle where a second lobe rises above threshold — where a grating lobe starts, not that everything past it is worse.



Scan loss

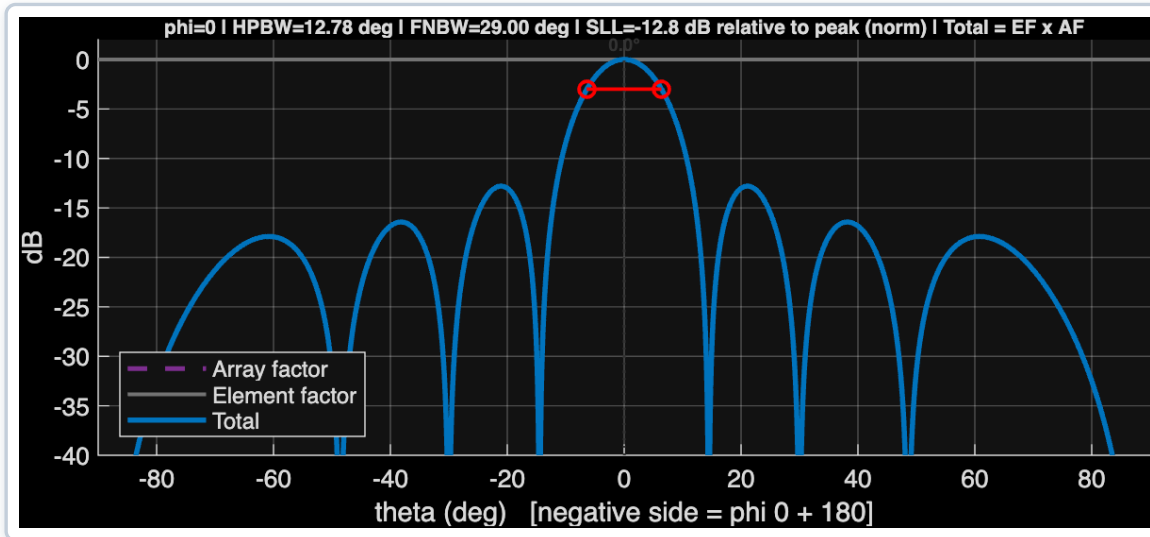
Plots gain relative to broadside against steer angle, next to the idealised $\cos(\theta)$ curve. The gap between them is your element pattern.

Phase scheme map

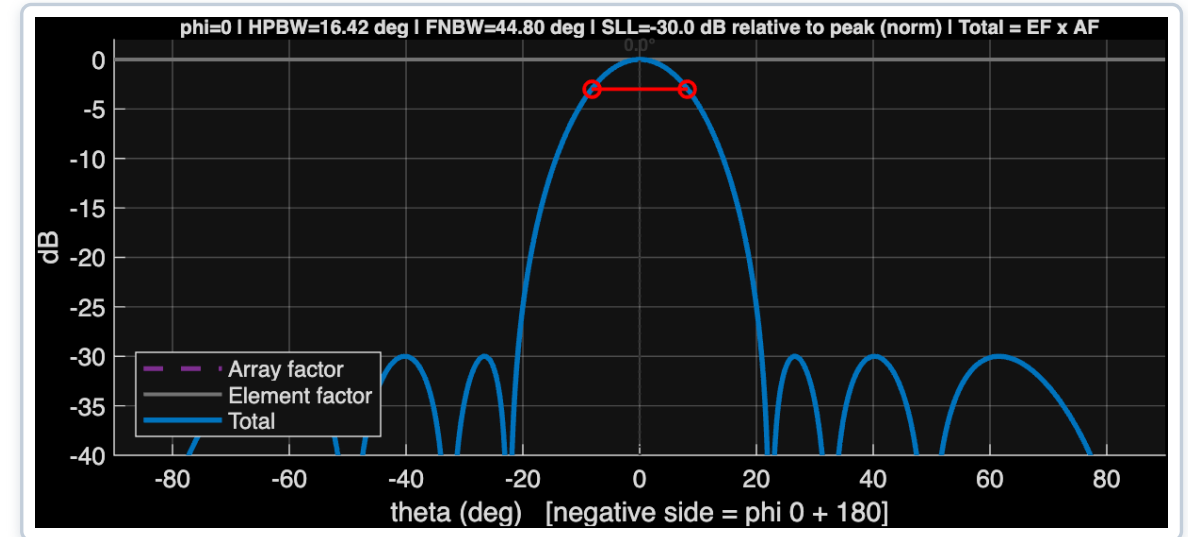
The per-port table: phase, amplitude, lattice position, which feed, and the frequency mode it was computed under. Has its own CSV and CST-macro exports.

Trading beamwidth for sidelobes

Same array, same steering. Chebyshev and Taylor are synthesised to a fixed 30 dB design sidelobe level.



Uniform — HPBW 12.78°, first sidelobe -12.8 dB



Chebyshev — HPBW 16.42°, sidelobes -30.0 dB

Uniform

Highest directivity. Sidelobes fixed at -13 dB whatever you do.

Chebyshev

Every sidelobe equal, at the design level. Widest of the classic three.

Taylor

Near-Chebyshev, but the sidelobes decay outward instead of staying level.

Manual (table)

Your own amplitudes. The taper code returns early and leaves them alone.

View tab — what is drawn, and how

3D surface shows

Total (EF × AF) · Array factor only · Element factor only · Axial Ratio (dB). The plot TITLE describes whatever is drawn — on "Element factor only" it reports the element's figures, not the array's.

Polarization readout

Total · RHCP · LHCP. On a circularly polarised array these are genuinely different patterns.

3D radius scale + Dynamic range

dB or linear radius, and how far down the dB surface is plotted (10–60 dB).

Absolute level / Full sphere

Real levels instead of peak-normalised, and theta 0–180° instead of the hemisphere. Both are DISPLAY only — neither changes a computed number.

The screenshot shows two panels of the control interface. The top panel has settings for '3D surface shows' (Total (EF x AF)), 'Polarization readout' (Total (any pol)), '3D radius scale' (dB (dynamic range)), and 'Dynamic range (dB)' (40). It also has checkboxes for 'Show absolute level in dBi' and 'Show full sphere (theta 0-180)'. The bottom panel has settings for 'Cut plane' (Theta cut (fixed phi_s)), 'Cut phi (deg)' (0), 'Quick cut plane' (phi=0, phi=90, Follow), a checked box for 'Cut plane follows steering phi', 'Theta for phi-cut (deg)' (0), and an unchecked box for 'Cut plot shows Axial Ratio (not gain)'. Blue circles and lines highlight specific settings and their corresponding descriptions in the surrounding text boxes.

Cut plane + Cut phi

Theta cut at fixed phi, or phi cut at fixed theta, and which plane the cut is taken in.

Quick cut plane

phi=0 / phi=90 / Follow. "Follow" keeps the cut on the beam as you steer, which is usually what you want.

Cut plot shows Axial Ratio

Swaps the cut from gain to axial ratio, for checking circular polarisation.

Reading the principal cut

Title readouts

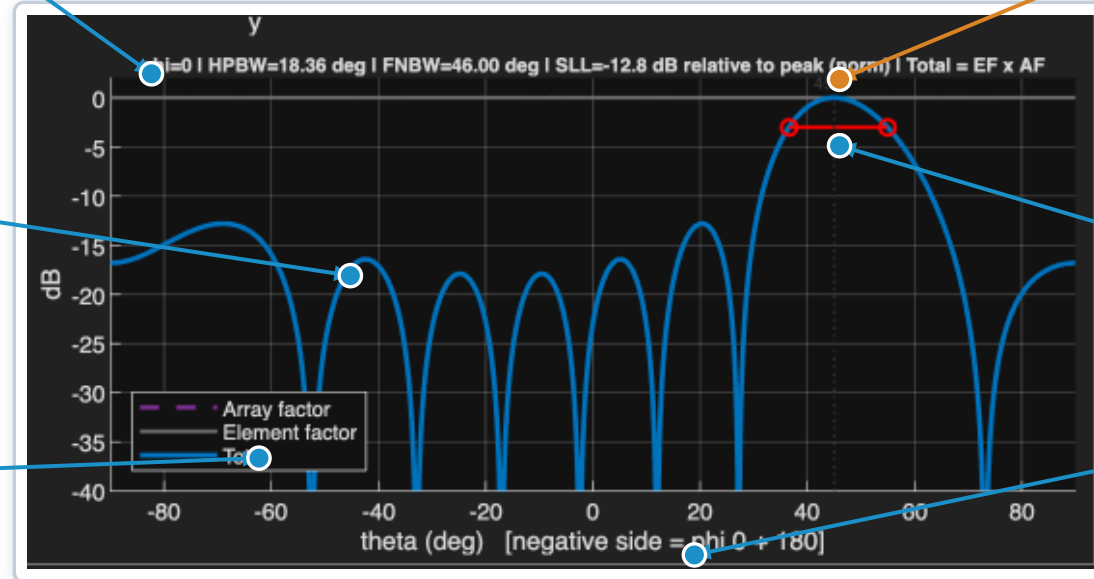
HPBW, first-null beamwidth and the highest sidelobe relative to the peak — recomputed on every change.

Sidelobes

Everything either side of the main beam. The title's SLL figure is the highest of them.

Three curves

Array factor (dashed), element factor (grey) and their product, Total (blue). Seeing all three shows which one is shaping the result.



Peak label

Where the beam ACTUALLY is. Under beam squint this reads e.g. "33.0° (steered 45°)" — the commanded angle in brackets.

Half-power markers

The two -3 dB points, drawn on the beam being measured.

Negative angles

The left half is phi + 180°, so one theta cut shows the whole plane rather than one side of it.

Reading the 3D pattern

What is drawn

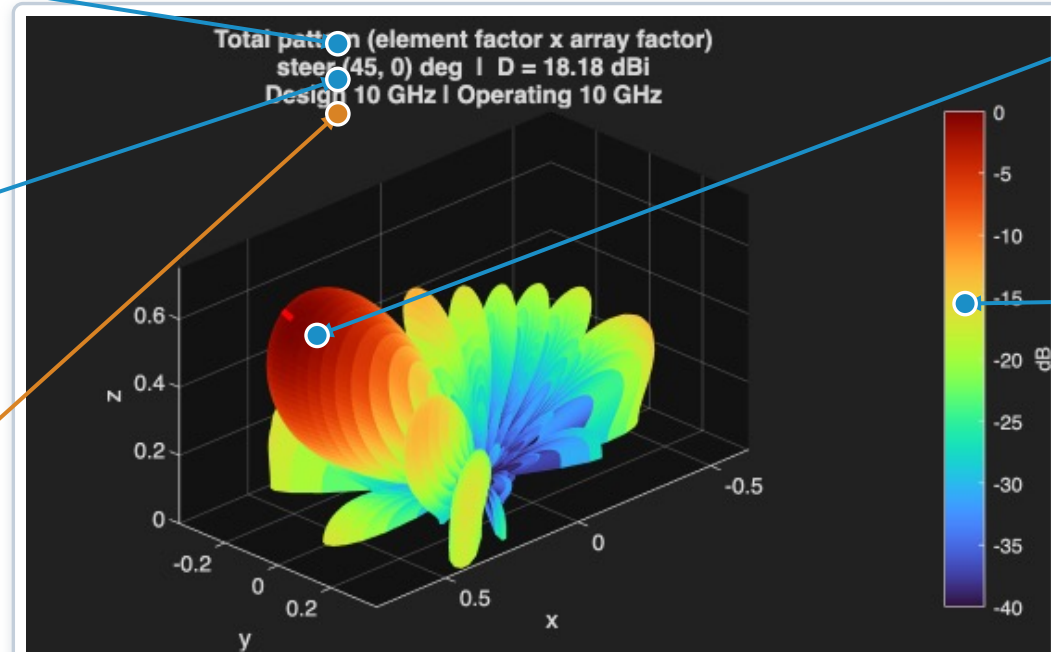
Follows "3D surface shows" on the View tab — total, array factor, element factor or axial ratio.

Steer angle and D

The commanded direction, and the directivity — or the GAIN, when the element came from an imported unit cell.

Frequency line

Design and operating frequency. With Beam squint on, this line also says where the beam actually landed, or that it left visible space.



The surface

Radius is level, colour is level. A red dot marks the peak direction.

Colour bar

dB below the peak, spanning the Dynamic range you set on the View tab.

Reading the Metrics panel

Elements / sum of amplitudes

How many elements are active, and their total weight.

Taper efficiency

The directivity given up to the amplitude weighting.

Frequency mode

RETUNED phases · PHASES FIXED · BEAM SQUINT — and under squint, the commanded angle next to the angle the beam actually reaches.

Span and spacing

The real aperture size and the smallest gap between two elements, in design wavelengths.

```

Elements: 64
Sum of amplitudes: 64.00
Taper efficiency: 1.000 (-0.00 dB loss)
Frequency mode: RETUNED phases | design 10 GHz |
operating 10 GHz | phase ref 10 GHz | steering phase retuned
for theta 45.00 deg, phi 0.00 deg
Element-center span: 3.50 x 3.50 λ (design)
Nearest-neighbour spacing: 0.5000 λ (design)
highest sidelobe: theta=69 deg, phi=180 deg, 12.8 dB below
peak
Area: 16.00 λ² (design)
Ideal directivity: 23.03 dBi
Calculated directivity: 18.18 dBi
Aperture efficiency: 32.7% (-4.85 dB)
    
```

Highest sidelobe

Its direction, and how far below the peak it sits — the same figure the cut plot's title reports.

Area / Ideal directivity

The aperture, and the $4\pi A/\lambda^2$ ceiling it could reach at best.

Calculated directivity

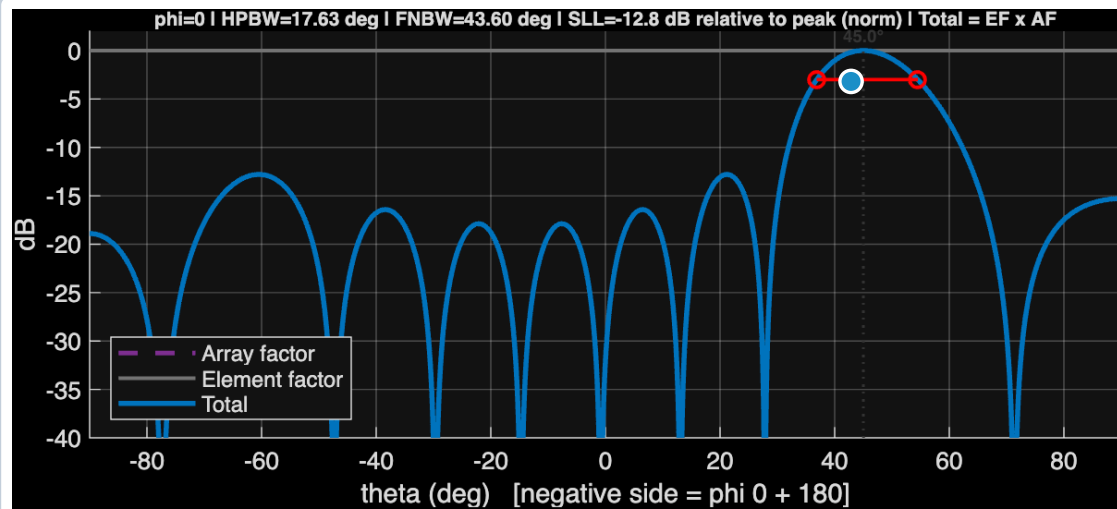
What the array actually achieves, from the pattern integral over the full sphere.

Aperture efficiency

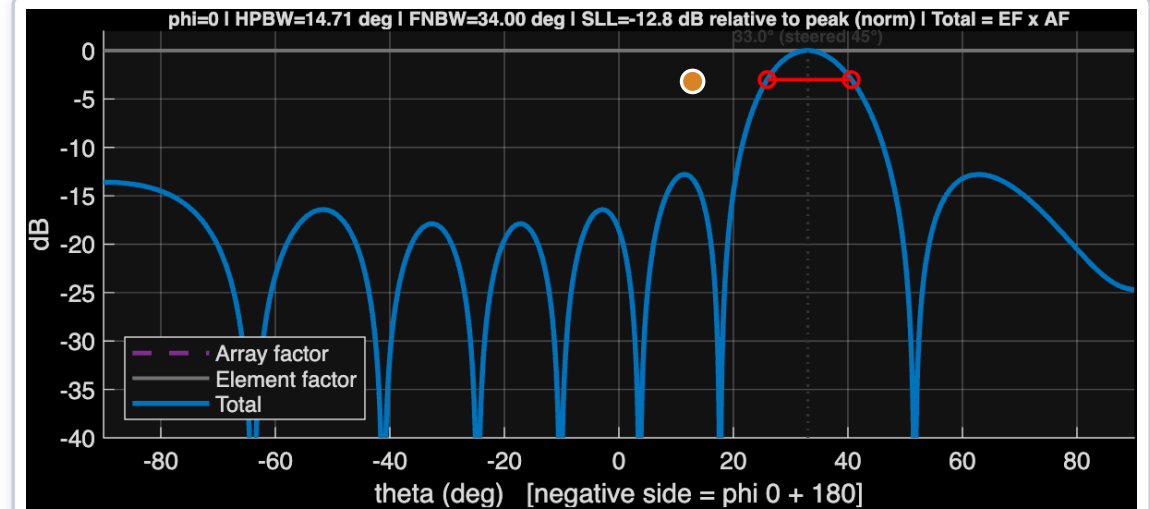
Calculated ÷ ideal. The headline number for how well the aperture is being used.

Beam squint — the same array, both ways

8 × 8 at 0.4λ , commanded to $\theta = 45^\circ$, design 10 GHz. The operating frequency is then raised to 13 GHz. Only the checkbox differs.



Squint OFF — beam holds at 45°



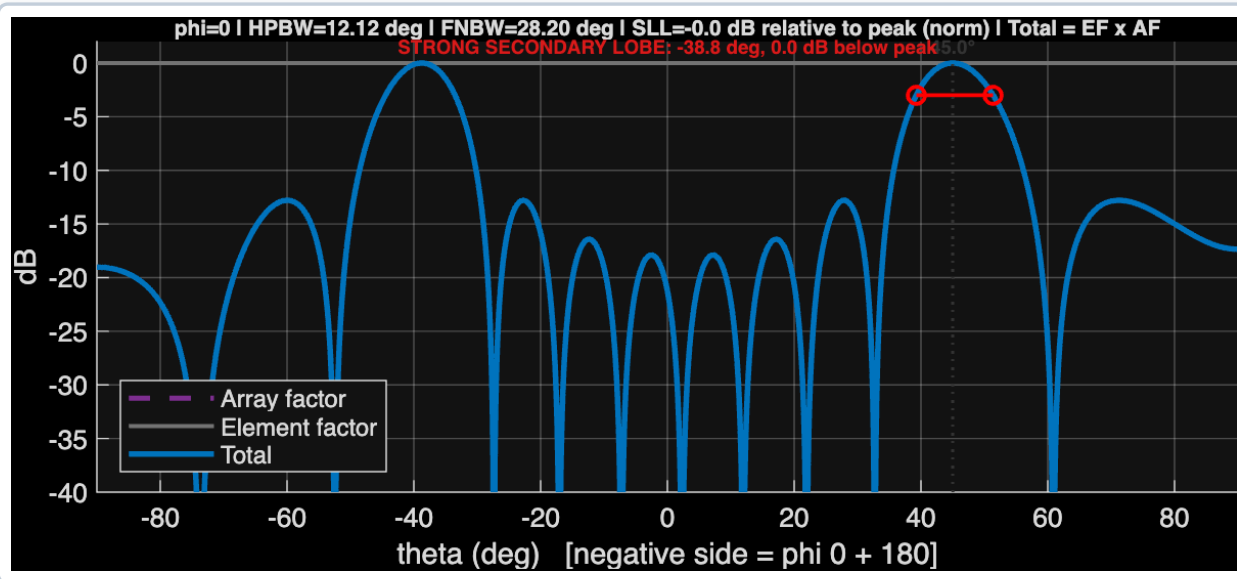
Squint ON — beam moved to 33°

$$\sin \theta_{\text{beam}} = \sin \theta_{\text{steer}} \times f_{\text{design}} / f_{\text{operating}}$$

$\text{asin}(\sin 45^\circ \times 10/13) = 32.95^\circ$, and the app puts the peak at 33.0° . Spacing was kept at 0.4λ on purpose: at 0.5λ this same frequency step also raises a grating lobe, which is a real effect but a different one.

Max scan — how far can this array steer?

Sweeps the steering angle in 2° steps and stops at the first angle where a second lobe rises above the detection threshold.



WHAT IT REPORTS BACK

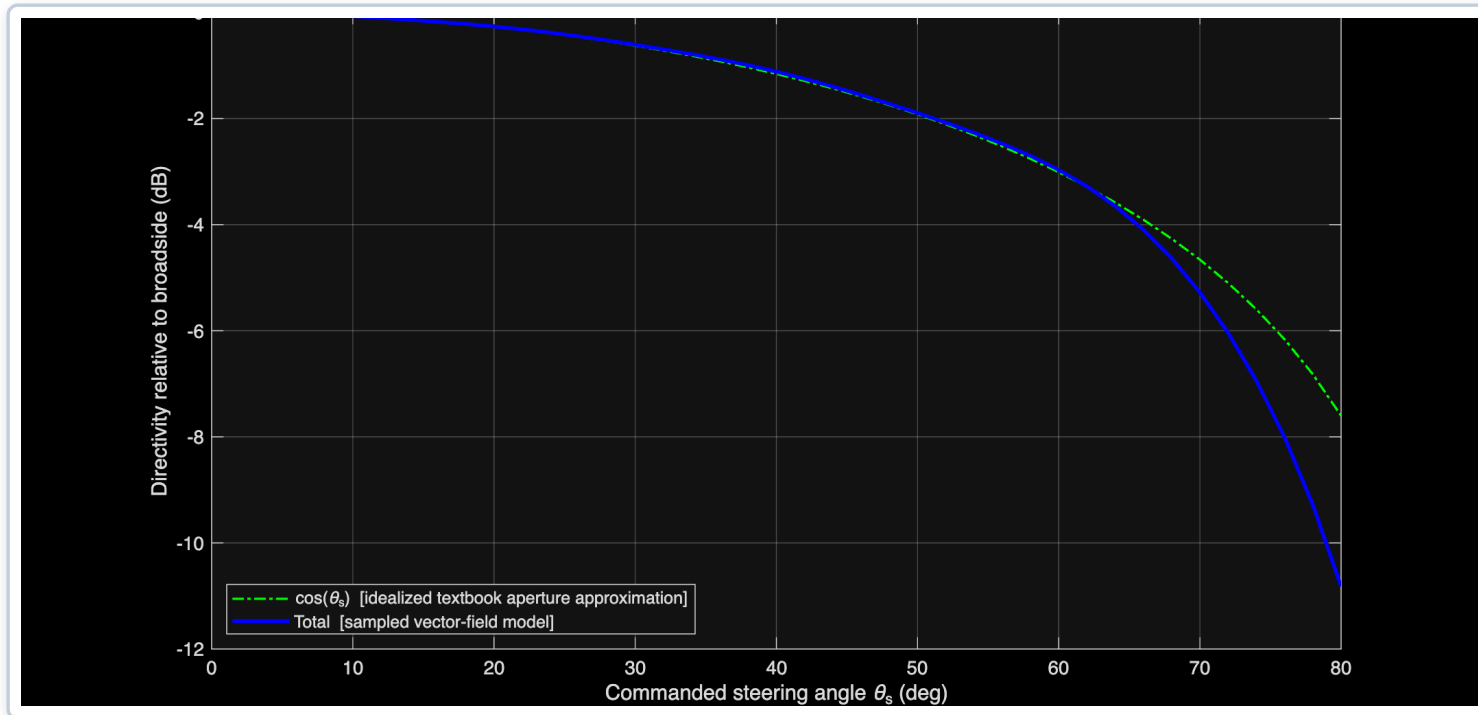
"STRONG SECONDARY LOBE detected at theta_s = 20 deg (phi_s = 0 deg held fixed). At 20 deg a repeat of the main beam sits at theta = 82 deg, phi = 180 deg, only 0.0 dB below the peak."

It is careful about what it claims: the sweep STOPS at the first crossing, so larger angles were never examined and are not implied to be worse. A lobe that only grazes the horizon is reported as grazing, not as a grating lobe.

The underlying condition

A grating lobe enters visible space when $\sin(\theta_s) \geq \lambda/d - 1$. At $d = 0.5 \lambda$ that never happens at any scan angle, which is exactly why half-wavelength spacing is the default choice. At 0.75λ it appears from about 19° onward.

Scan loss — what steering costs you



Plotted relative to broadside, so it answers "how much do I lose by scanning here" — not "what is my gain"

$\cos(\theta_s)$

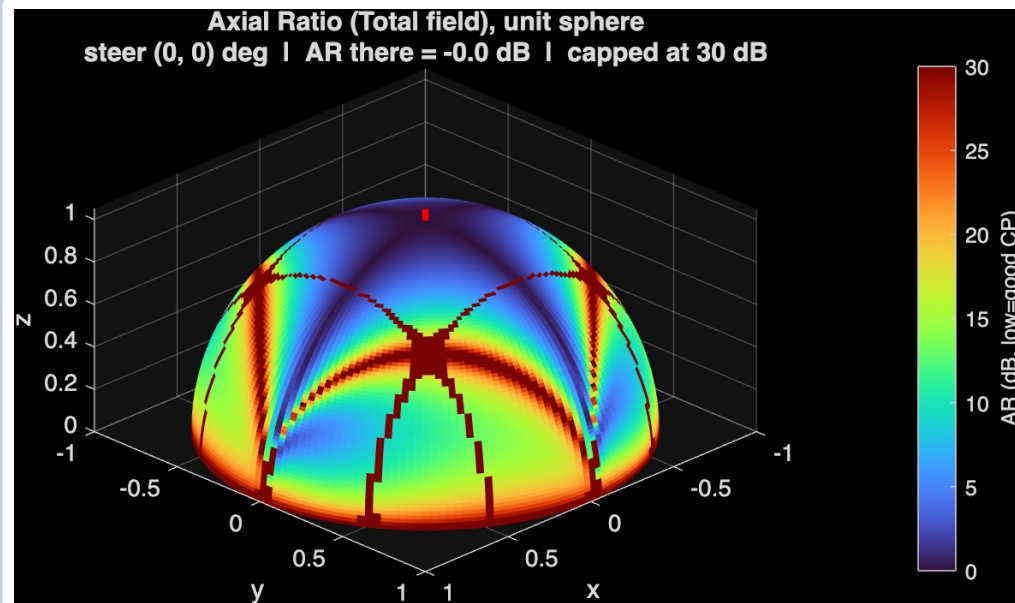
The idealised textbook approximation — projected-aperture loss and nothing else. Dash-dot green.

Total

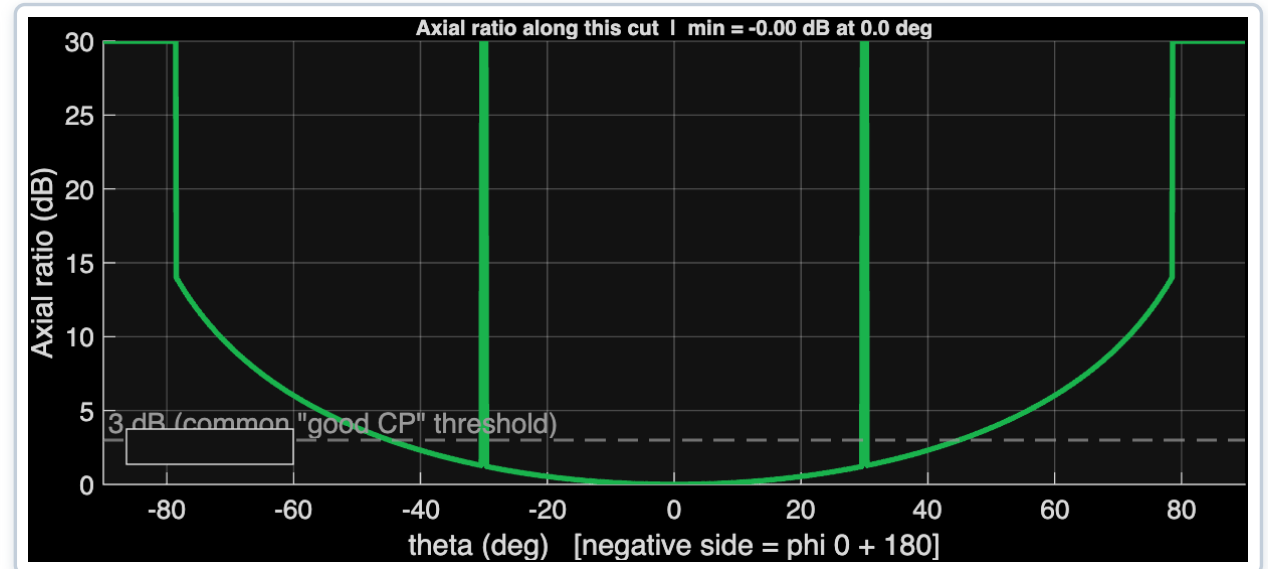
The sampled vector-field model: your actual elements, spacing, taper and rotation. This is the one to believe. The gap between the two curves IS your element pattern and lattice showing up — a directive element pulls the blue curve well below cosine.

Polarisation and circular quality

The pattern is computed as a vector field — theta and phi components summed separately, magnitude taken only at the end. That is what makes the polarisation readouts meaningful rather than cosmetic.



Axial ratio over the sphere — blue (low) is good CP



The same thing as a cut, with "Cut plot shows Axial Ratio" ticked

Polarisation readout

Total · RHCP · LHCP. On a circularly polarised array these are genuinely different patterns, not rescaled copies of one another.

Sequential rotation

Rotate elements physically over a repeating block and add the matching feed phase. The compensating sign follows the element's MEASURED handedness, not an assumption.

Magnitude-only imports

cannot carry polarisation. Mix one with rotated elements and the plot says so rather than inventing a phase.

The Phase scheme map — numbers for hardware or CST

Single feed | 4 × 4 | commanded θ 30° | ϕ 0°
 RETUNED phases @ 10 GHz | operating 10 GHz | steering phase retuned for commanded angle
 Single-feed map | H/V feed-side correction is not used | layout editor currently Default: check your CST port layout

Read the header before you trust the numbers

Scheme · array size · commanded angle, then the frequency the phases are referenced to and whether beam squint was on.

Feed scheme

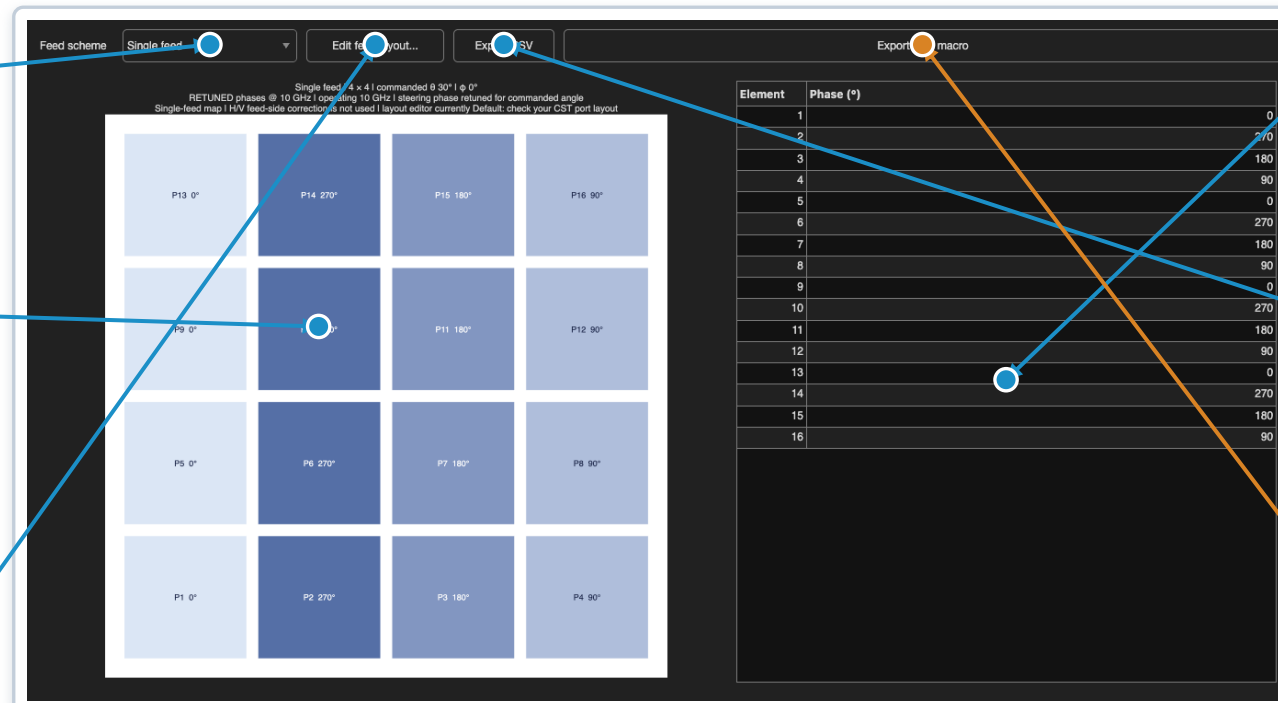
Single feed · Dual feed LP · Dual feed CP.
 Sets how many ports each element has and what phase each one gets.

Port tiles

One tile per port, labelled P<n> with its phase. Colour tracks phase, so a scheme's periodic structure shows at a glance.

Edit feed layout...

Opens the repeating-unit editor — next slide.



Per-element table

The same phases as plain numbers, ready to read off or copy.

Export CSV

phase_scheme.csv — one row per port with phase, amplitude, lattice position and the frequency mode.

Export CST macro

A .bas VBA macro defining the port excitation (CombineResults). A definition only — it never starts the solver.

The feed layout editor — declaring the repeating unit

Repeat every (rows) / (columns)

The period of the unit. A lattice cell maps to a row of the table as $\text{mod}(\text{row}-1, \text{nRows})+1$, $\text{mod}(\text{col}-1, \text{nCols})+1$.

Block_Row / Block_Col

Which cell of the repeating unit this row describes. Cell (1,1) is the bottom-left element.

H_port

Which of an element's two ports carries H: odd ($2n-1$) or even ($2n$).

One row per cell of the unit that repeats across the array. Lattice row/col map to a cell as $\text{mod}(\text{row}-1, \text{nRows})+1$, $\text{mod}(\text{col}-1, \text{nCols})+1$ for whatever period you set above, so cell (1,1) is the bottom-left element.

H port: which of an element's two ports (odd = $2n-1$, even = $2n$) carries H. H side / V side: which edge that feed sits on. These are read off your CST model - the tool cannot derive them, because a mirrored element has no rotation angle to give it away.

Repeat every (rows)

and (columns)

2

Block_Row	Block_Col	H_port	H_side	V_side
1	1	1 odd ($2n-1$)	left	bottom
2	1	1 odd ($2n-1$)	left	top
1	2	2 odd ($2n-1$)	right	bottom
2	2	2 odd ($2n-1$)	right	top

Cancel

Apply

H_side

Which edge that feed sits on — left or right.

V_side

The same for the vertical feed — top or bottom.

Apply

Writes the layout back; Cancel discards it.

Why this is declared and not inferred

Feed sides are read off your CST model — the tool cannot derive them. A mirrored element has no rotation to give it away, so two elements sharing a block column can still feed from opposite sides. Guessing would silently flip a polarisation.

If the tool refuses something

Every refusal names its cause. These are the ones you are most likely to meet.

"Only realized-gain dBi exports ... are supported"

Your CST export is directivity, plain gain, a linear E-field, or Ludwig-3 co/cross. Re-export as realized gain in dBi with Theta/Phi components.

"A complete regular Theta/Phi grid is required"

The export has missing angles or is folded (negative theta). Re-export theta 0...180° with a full phi circle.

"Export realized gain with Abs(total), Abs(Theta)..."

The header does not carry the five expected columns. Check nothing extra was included in the export.

"This design needs its imported far-field pattern"

A loaded config expects an imported element that is not present. Load the pattern, or pick another element factor.

"Scan loss is defined as a RATIO to the broadside beam"

Asked for scan loss where broadside gives nothing to compare against. Use a case with a real broadside beam.

"single element" / "Theta cut" refusals

An analysis tool was asked for something it cannot answer in the current mode — add elements, or switch the cut plane.

Reported gain does not move with frequency

Expected, for an imported unit cell. The pattern was characterised at one frequency, so the tool will not extrapolate it to another.

Quick reference

Rules of thumb

- $dx = 0.5 \lambda$ — no grating lobe at any scan angle.
- Uniform taper — sidelobes fixed at -13 dB.
- $\cos^q$ element — $D = 2(q+1)$.
- Doubling the element count adds 3 dB.

Easy mistakes

- Reading the steer box as the beam angle while Beam squint is on.
- Exporting TSV with the wrong Design frequency — every millimetre scales with it.
- Leaving "3D surface shows" on Element factor only, then reading the title as an array figure.
- Exporting directivity from CST instead of realized gain.

Where to look

- Beam direction — the cut plot's peak label.
- Frequency mode — Metrics panel, near the top.
- Scan limit — Max scan.
- Loss vs angle — Scan loss.
- Port phases — Phase scheme map.

Aperture ceiling

$$D_{ideal} = 4\pi A / \lambda^2$$

Beam squint

$$\sin \theta_b = \sin \theta_s \cdot f_d / f_{op}$$

Grating lobe onset

$$\sin \theta_s \geq \lambda / d - 1$$

TSV position

$$X(m) = x(\lambda) \cdot \lambda_{design}$$

Every screenshot and every arrow position was taken from the running app.